The Threat of Overfishing on Biodiversity and the Environment: An Examination of Global Efforts and the Need for Continued Action

ABSTRACT

Overfishing is an urgent problem that needs to be resolved. While there have been measures to tackle overfishing such as having marine protected areas and imposing limits on fishing, the efficiency of these measures is disputable. Hence, this project aims to provide data driven suggestions on what more can be done to address overfishing to protect our marine biodiversity and environment through (a) understanding the current state of seafood & fish production and consumption (b) analyzing the impact of overfishing (c) assessing existing global efforts to mitigate overfishing – its' limitations and opportunities.

PROBLEM STATEMENT AND MOTIVATION

Overfishing poses a significant threat to food security. Globally, over 3 billion people depend heavily on fish as their main source of protein (Future of Fish, n.d.). As overfishing destroys marine life (including fish supply), this leads to a shortage of food in underdeveloped countries where switching to alternatives is difficult due to access and financial constraints (Leschin-Hoar, 2016).

Additionally, overfishing harms the environment. It reduces the ability of the ocean, the largest carbon sink in the world, to effectively absorb carbon dioxide from the atmosphere (United Nations Climate Change, 2022). This means that more carbon dioxide, a greenhouse gas, gets trapped in the air accelerating global warming. Global warming in turn puts us at risk when temperatures become too hot to survive.

Threatened food security and climate change are just two consequences of overfishing. Given the severity and the widespread impact of overfishing, it is thus crucial that this issue is addressed.

RELATED WORKS

This section explores the existing data and visualizations on the topic of overfishing. However, there are limitations in the current approaches which our team aims to refine in our project to generate more meaningful insights.

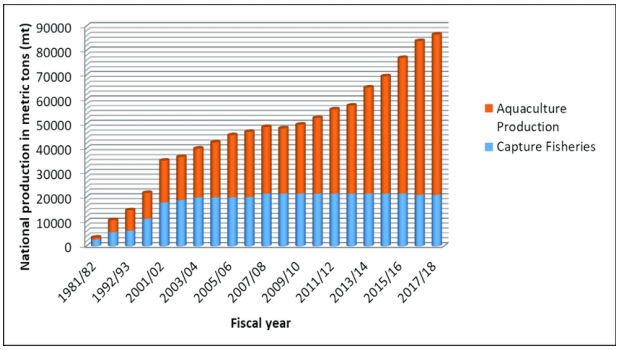


Figure 1. Citation: South Asian Association for Regional Cooperation

1. Fish Production trends

Figure 1 illustrates the production of fish over the past 36 years. This visualisation has limitations on "where" are the fishes being produced, and how aquaculture production compare with wild fish catch. Our team would utilize this data to assess the sustainability of aquaculture in relation to wild fish catch.

2. Overfishing over the years

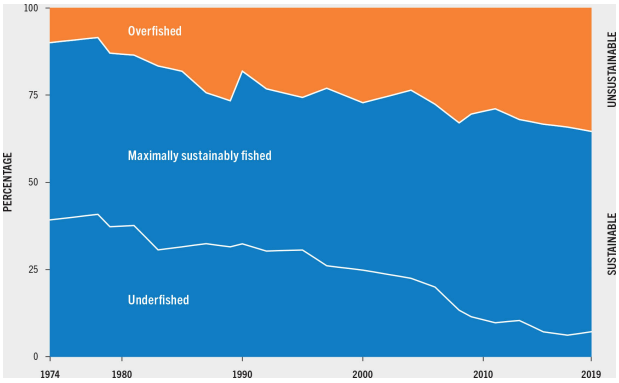


Figure 2. Citation: Food and Agriculture Organisation of UN

Figure 2 illustrates the global trends in the state of the world's marine fishery stocks from 1974 to 2019. There is an increase in unsustainable fishing practices and the share of overfished

stocks increased from approximately 10% in 1974 to 1/3 of the world's fish stocks in 2019. However, due to the static nature of this chart and the lack of labels, we are unable to tell if overexploitation is progressively getting worse or better over time in relation to the biologically sustainable fish stocks. Moreover, the shading under the boxes also makes the chart confusing because we are unable to tell what the significance of the area was in this chart.

3. Human impact on Marine Ecosystems

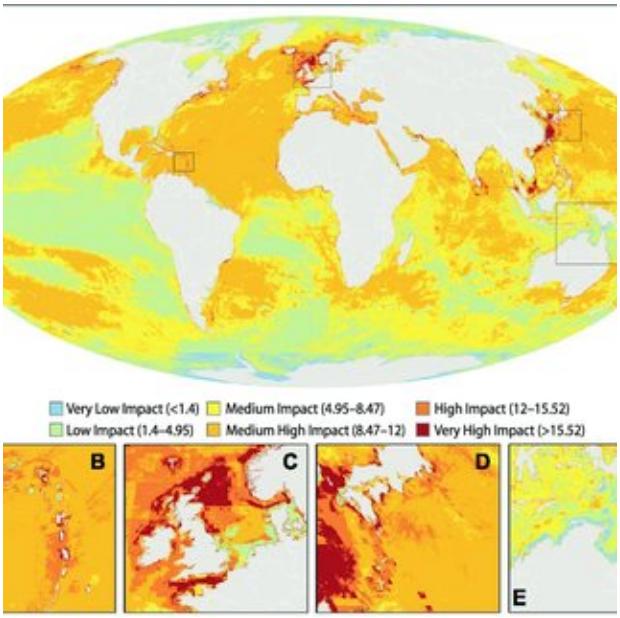


Figure 3. Citation: A Global Map of Human Impact on Marine Ecosystems

Figure 3 is part of a research that shows human activities have significantly impacted the majority of the world's oceans, with only a small percentage of marine ecosystems remaining relatively untouched. Fishing was found to be the most widespread and intense human impact on marine ecosystems, with over 40% of the world's oceans subject to high levels of fishing pressure. This highlights the urgent need for improved management of fisheries and the development of more sustainable practices in industries that operate in or near the ocean, in order to reduce negative impacts and preserve biodiversity in the world's oceans. Our team wants to capitalise on this research to show the extent and magnitude of human impact on the world's oceans, and to highlight areas that are most at risk due to overfishing.

4. Protected Marine Territorial Waters

Figure 4 maps the areas of the chondrichthyan fish species that are threatened. This includes sharks and rays but does not cover all types of generic fishes. Our team aim to explore the effectiveness of Marine Protected Areas (MAPs) by investigating the relationship between the percentage of protected marine areas and the severity of overfishing across all types of marine species in a particular country.

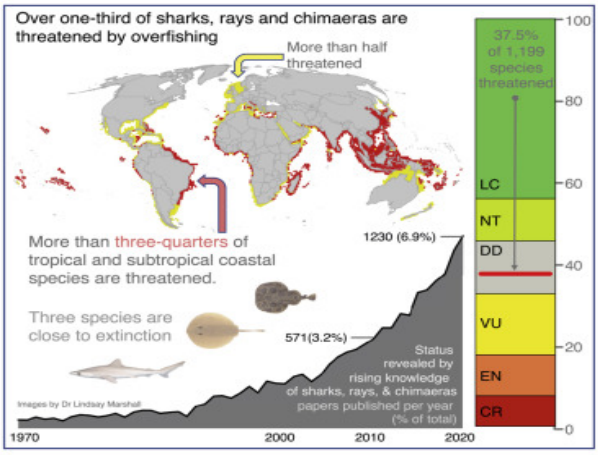


Figure 4. Citation: 1/3 chondrichthyan fish species are threatened

DATASETS

In this project, we tentatively plan to use 10 datasets in total to explore the problem and consequences of overfishing. These datasets span across a wide variety of domains, including seafood production/consumption trends, commercial fishing methods, environmental impact of fishing, trends in fish farming, and fishing wastage and bycatch. The details of each dataset are elaborated in Table 1.

Dataset	Source
Global seafood production	Food & Agriculture - UN
Fish farming production	World Development Indicators - World Bank
Methods of wild fish catch	United Nations FishStat Database
Sustainability of fish stocks	Food and Agriculture Organization of the United Nations
Fishing intensity by region	RAM Legacy Stock Assessment Database
Fish stocks and maximum sustainable yield	RAM Legacy Stock Assessment Database
Environmental impact of farmed seafood	Environmental performance of blue foods
Fish wastage from type of fishing gear	A third assessment of global marine fisheries discards
Distribution of ocean areas protected from fishing	Protected Planet: The World Database on Protected Areas (WDPA) and World Database on Other Effective Area-based Conservation Measures (WD-OECM)
Distribution of protected territorial waters	World Development Indicators - World Bank

Table 1. Datasets and their sources

However as the project progresses, we may elect to source for more datasets or remove some of our existing ones.

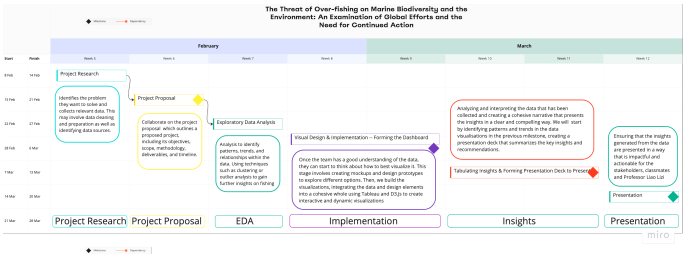


Figure 5. Project Milestones gantt chart

TOOLS AND RESOURCES USED

A. Data Preparation

Before making any assumption with the datasets, we will perform Exploratory Data Analysis (EDA) to identify any trends and gain insights within the data. With a better understanding of the datasets, we can drop unnecessary columns and perform data cleaning using Python by importing packages such as Pandas and NumPy, to remove missing values and outliers. Our team may also convert categorical columns into dummy variables and join datasets if necessary.

B. Visualization

Our team will utilize Tableau to create most of our visualizations and D3.js for specific dynamic visualizations. The goal is to leverage these tools to create custom visualizations that are tailored to the specific needs of our project, so that we can derive meaningful insights which will shape our recommendations.

PRELIMINARY RESULTS

From the 10 datasets we have retrieved, the team performed preliminary analysis to better understand the data using visualisations.

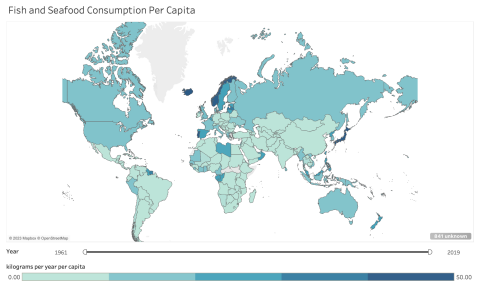


Figure 6. Fish Seafood Consumption around the world

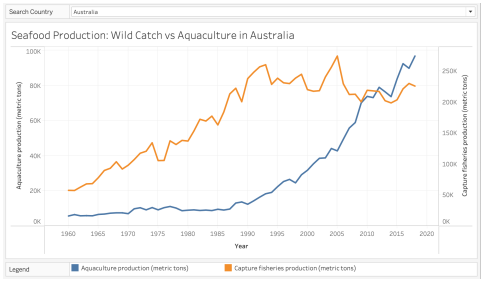


Figure 7. Seafood Production over the past 50 years

From our preliminary visualisation in Figure 6, the team identified that certain countries like Iceland has a significantly higher fish and seafood consumption compared to the rest of the world. It would be interesting to understand how overfishing rates are like Iceland as part of our project. From Figure 7, it appears that aquaculture is a sustainable alternative to wild catch fisheries because the seafood production from aquaculture has rapidly increased, even exceeding that of wild catch fisheries in between 2010 - 2015.

PROJECT MILESTONES

The team will began by identifying the problem we want to solve. This step involves extensive research and analysis to understand the business problem or challenge and its underlying causes. For this project, the problem of overfishing was identified. Subsequently, the team will move on to gather relevant data from various sources, which may include public datasets, surveys, or internal data from the organization. Then, the team will proceed to clean the data before it is used for analysis. This may involve removing duplicates, fixing formatting errors, and filling in missing data points. This is to ensure that the data collected is accurate, reliable, and relevant to the problem that we are trying to solve. Once the data has been preprocessed, the team will start identifying patterns and trends in the data which will drive meaningful insights to aid in our recommendations.

Refer to Figure 5 for the timeline of the project.

REFERENCES

[1] Fish is Food (n.d.), *Fish is Food: feeding people, nourishing communities.*, `http://futureoffish.com/blog/fish-food-feeding-people-nourishing-communities#:~:text=Globally%2C%20over%203%20billion%20people,depleted%20the%20ocean's%20natural%20resources`

[2] Leschin-Hoar, C. (2016, June 30)., *1 In 10 People May Face Malnutrition As Fish Catches Decline.* , `https://www.npr.org/sections/thesalt/2016/06/30/484015370/1-in-10-people-may-face-malnutrition-as-fish-catches-decline`

[3] United Nations Climate Change. (2022, June 10), *Plenty of Fish?*, `https://unfccc.int/blog/plenty-of-fish#:~:text=%E2%80%9COverfishing%20and%20destructive%20fishing%20not,is%20needed%20for%20climate%20mitigation`